

Solution for Module 1, Intro to Stats

classmate
Date
Page

Q1.) Sol:-

H_0 = null hypotheses

no difference in intelligence of boys & girls

$$\text{i.e., } \mu_1 = \mu_2$$

H_1 = alternative hypotheses

different in terms of intelligence

$$\text{i.e., } \mu_1 \neq \mu_2$$

Given,

$$\bar{x}_1 = 89 \text{ (mean}_{\text{girls}})$$

$$\bar{x}_2 = 82 \text{ (mean}_{\text{boys}})$$

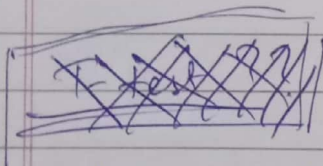
$$\text{S.D.}_1 (\sigma_1) = 4$$

$$\text{S.D.}_2 (\sigma_2) = 9$$

$$\text{sample size } (n_1) = 50$$

$$\text{" " } (n_2) = 120$$

$$\text{L.O.S (level of significance)} = 5\%$$



Using pooled t-test,

$$t = (\bar{x}_1 - \bar{x}_2) / (\text{s.d} * \sqrt{(1/n_1 + 1/n_2)})$$

Calculating pooled s.d.,

$$\sqrt{\frac{(\sigma_1^2 (n_1 - 1) + \sigma_2^2 (n_2 - 1))}{(n_1 + n_2 - 2)}}$$

Where, S_1 = S.d of sample of girls.

S_2 = " " " " " boys.

$$\therefore \text{Pooled S.D} = \sqrt{\frac{(14^2(50-1) + 9^2(120-1))}{(50+120-2)}}$$

$$= \sqrt{\frac{16 \times 49 + 81 \times 119}{168}}$$

$$= \sqrt{\frac{784 + 9639}{168}}$$

$$= \sqrt{\frac{10423}{168}}$$

$$\approx 7.86.$$

Calculating t -statistic,

$$t = (89 - 82) / (7.86 * \sqrt{1/50 + 1/120})$$

$$\approx 3.68.$$

$$df(\text{degree of freedom}) = n_1 + n_2 - 2$$

$$= 50 + 120 - 2$$

$$= 168$$

$$p = 2 \times (1 - 0.99985)$$

$$= 2 \times 0.00015$$

$$= 0.0003.$$

Since, p value is less than level of significance i.e. 0.05, we reject the null hypothesis.

Q2. Sol:-

Category	Diagnosed as Cancer	Without Cancer	Total
Smokers	220	230	550
Non-smokers	350	640	990
Total	680	910	1590

~~Expected value = row total~~

H_0 (null hypothesis): Smoking and cancer are independent.

H_1 (alternative hypothesis): Smoking and cancer are dependent.

Using l.o.s as 0.05 (5%)

Expected value for smokers diagnosed with cancer:

$$E = \frac{550 \times 680}{1590} = 235.22$$

Expected value for smokers without cancer

$$E = \frac{550 \times 910}{1590} = 314.78$$

Expected value for non smokers ~~diag~~ diagnosed with cancer :

$$E = \frac{990 \times 680}{1590} = 444.78$$

Expected value for non smokers without cancer :

$$E = \frac{990 \times 910}{1590} = 575.22$$

So, we have our expected value as :

Category	Diagnosed as cancer	Without cancer
Smokers	235.22	314.78
Non-smokers	444.78	575.22

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \text{where, } O = \text{observed value}$$

$E = \text{Expected value}$
 $\chi^2 = \text{chi-square value}$

For smokers ~~diag~~ diagnosed with cancer,

$$\chi^2 = \frac{(220 - 235.22)^2}{235.22} = \frac{(-15.22)^2}{235.22} = \frac{231.65}{235.22} = 0.985$$

For smokers without cancer,

$$\chi^2 = \frac{(230 - 314.78)^2}{314.78} = \frac{(-84.78)^2}{314.78} = \frac{7187.17}{314.78} = 22.83$$

For non-smokers diagnosed with cancer,

$$\chi^2 = \frac{(350 - 444.78)^2}{444.78} = \frac{(-94.78)^2}{444.78} = \frac{8982.22}{444.78} = 20.19$$

For non-smokers without cancer,

$$\chi^2 = \frac{(640 - 575.22)^2}{575.22} = \frac{64.78^2}{575.22} = \frac{4196.61}{575.22} = 7.29$$

Total chi-square value,

$$\chi^2 = 0.985 + 22.83 + 20.19 + 7.29 = 51.30$$

$df = (r-1) \times (c-1)$ where, r = no. of rows
 c = no. of columns

$$df = (2-1) \times (2-1) = 1$$

~~Crit~~ Critical value for $df = 1$, $\alpha = 0.05$,
 $= 3.841$

Since calculated $\chi^2 >$ critical value,
we reject the null hypothesis.

Conclusion - The data suggest smoking and cancer are not independent. Smoking may increase the likelihood of cancer.